

Dual-Axis Gantry with 8-Channel Pipette *Automated Vertical & Horizontal Motion System for Biomedical Applications*

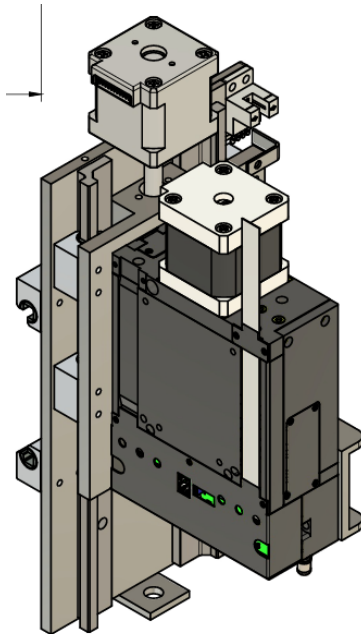
Project Overview: - Designed for precise movement of an 8-channel pipette. - Enables both vertical (Z-axis) and horizontal (X-axis) motion. - V-belt driven mechanism for smooth and efficient motion. - Suitable for integration in RNA/DNA extraction or PCR systems. - Compact and modular design for easy maintenance and upgrades. - High accuracy and stability during operation. - Low-friction movement ensures smooth and efficient motion.

Technical Specifications: -

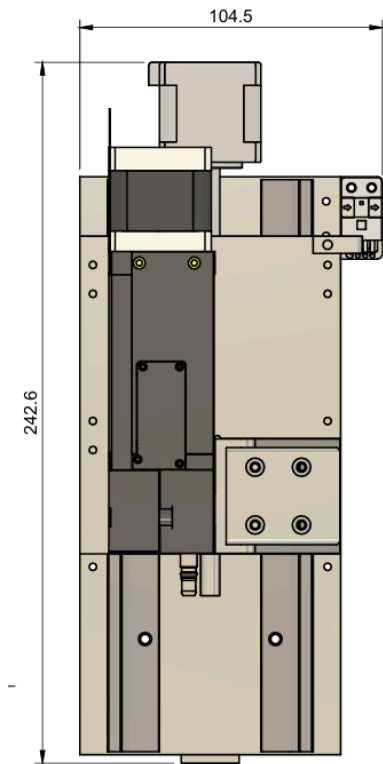
- **Vertical Motion:** Moons NEMA 17 Stepper Motor
- **Vertical Travel Mechanism:** Supported on single Igus slider for precise linear motion.
- **Horizontal Motion:** 4 Igus sliders on base plate for stable travel
- **Pipette:** 8-channel Keyto for high-throughput operations
- **Frame Material:** Aluminum (lightweight & rigid)
- **Travel Range:** - Vertical: 100 mm - Horizontal: can be adjusted
- **Couplers / Attachments:** Custom machined for precise alignment
- **Maintenance:** Easy replacement and upgrade of components
- **Drive Type:** V-belt driven mechanism for controlled motion

CAD Representation: -

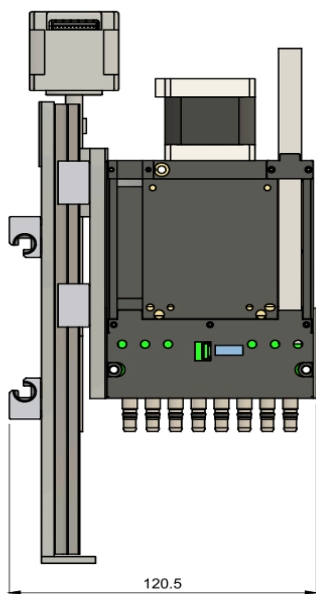
- **Isometric View:** Complete assembly showing full mechanism.



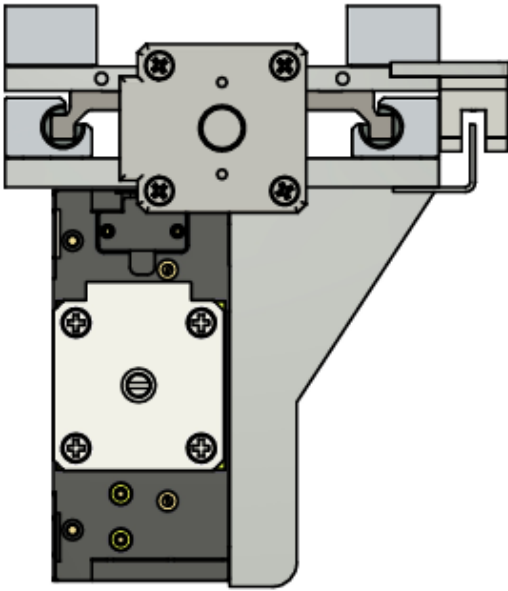
- **Front View:** Shows vertical movement of the pipette assembly.



- **Side View:** Shows horizontal travel along the sliders.



- **Top View:** Layout of base plate and horizontal sliders.



Design Highlights:

- **Vertical Motion:** - Single stepper motor drives pipette smoothly. - Slider ensures low-friction and precise movement. - V-belt transmission for smooth and reliable motion.
- **Horizontal Motion:** - Four Igus sliders provide stable horizontal travel. - Maintains alignment and minimizes vibration.
- **Modularity & Maintenance:** - Pipette assembly can be easily replaced or upgraded. - Allows integration into larger automated systems.
- **Performance:** - Designed for accurate, repeatable motion. - Ensures reliable operation for biomedical applications.

Materials & Components:

- **Stepper Motor:** Moons NEMA 17 dual shaft
- **Linear Guides:** Igus sliders for smooth, low-friction motion
- **Frame:** Aluminum for lightweight rigidity
- **Pipette:** 8-channel Keyto for high-throughput operations
- **Couplers / Attachments:** Custom for precise fit and alignment
- **Drive Mechanism:** V-belt system for controlled motion

Challenges Faced & Solutions: -

Challenge 1: Maintaining smooth vertical motion with minimal vibration.

Solution: Selected high-quality Igus slider and optimized mounting alignment.

Challenge 2: Ensuring stable horizontal travel without misalignment.

Solution: Used four evenly spaced Igus sliders and tested alignment repeatedly.

Challenge 3: Integrating stepper motor with pipette assembly.

Solution: Custom machined couplers and mounts for precise connection.

Challenge 4: Limited space for assembly on base plate.

Solution: Designed compact modular layout and optimized component placement. -

Challenge 5: Transmitting motion reliably without backlash.

Solution: Implemented a V-belt driven mechanism for smooth power transfer.

